

MeshNode System Manual

Multi-Device Setup, Ring All-Reduce AI Inference & Operating Guide

1. Executive Overview

MeshNode is a high-performance, decentralized peer-to-peer AI inference engine. It seamlessly connects disparate edge devices (Windows Desktop, iPad/iOS, and Android) into a cooperative **Ring All-Reduce** compute cluster. Utilizing zero-copy Rust FFI bindings, ONNX Runtime acceleration, and automated Zero-Knowledge (ZK) execution proofs, MeshNode allows devices to partition AI workloads safely across local networks.

2. Official Release Binaries

Platform	Artifact File	Size	Target Environment
iOS / iPad	Mesh_UI_Release.ipa	16.16 MB	iPadOS / iOS 15.0+ (Sideload / AltStore)
Android UI	mesh_ui_android_release.apk	16.34 MB	Android 8.0+ (ARM64 / Flutter UI)
Android Core	mesh_node_arm64.apk	16.26 MB	Android Native Background Daemon
Windows UI	mesh_ui_windows_release.zip	20.73 MB	Windows 10 / 11 (x64 Desktop Flutter)
Windows Node	mesh_node_windows_bundle.zip	19.67 MB	Windows CLI & ONNX Background Engine

3. Multi-Device Walkthrough (2-Node Ring Setup)

To connect two physical devices (e.g. **Device 1: Windows PC** and **Device 2: iPad / Android**), both devices must be on the same local Wi-Fi / LAN network. Follow these exact settings:

Setting Parameter	Device 1 (Windows PC)	Device 2 (iPad / 2nd Device)
Node ID	Node-A	Node-B
Listen Port	50061	50062
Next Peer Target	50062 (or Device 2 IP:50062)	50061 (or 172.20.10.4:50061)
Ring Size	2	2
Step 1 Action	Click 'Start Node'	Click 'Start Node'
Step 2 Action	Click 'Connect to Next Peer'	Click 'Connect to Next Peer'

Step-by-Step Execution Sequence:

- **1. Launch Applications:** Open MeshNode on both devices.
- **2. Start Node-A (Device 1):** Verify Port 50061, Target 50062, Ring Size 2. Press '**Start Node**'.
- **3. Start Node-B (Device 2):** Verify Port 50062, Target 50061, Ring Size 2. Press '**Start Node**'.
- **4. Establish Ring Link:** Click '**Connect to Next Peer**' on both devices. The libp2p mDNS service automatically verifies network reachability.

- **5. Execute Decentralized Inference:** In the bottom chat box on either device, enter a prompt (e.g. *'Run distributed activation test'*) and click **Send**.

Inference Pipeline Flow:

- 1. Originating node tokenizes the prompt and executes Layer 1-N ONNX forward passes.*
- 2. A Zero-Knowledge proof ($W \cdot X + B$) is mathematically synthesized over the intermediate tensor.*
- 3. The tensor payload is transmitted over gRPC to Peer 2 for subsequent layer execution.*
- 4. Peer 2 cryptographically verifies the ZK proof, computes its portion, and returns the accumulated result.*
- 5. The final ZK-Verified output is streamed and displayed in the real-time UI feed on both screens.*

4. Core Architecture & Features

A. Ring All-Reduce Topology

Unlike centralized client-server AI hubs, MeshNode uses a distributed Ring topology. Bandwidth demands remain flat regardless of total network nodes, as each node only transmits intermediate tensors to its immediate topological neighbor.

B. Zero-Copy Rust FFI (flutter_rust_bridge v2)

Heavy tensor manipulation, network encryption, and model inference take place in compiled native Rust. The Flutter UI accesses tensors with zero-copy shared memory pointers, ensuring zero UI jank and optimal 60/120 FPS performance.

C. Real-Time Telemetry & Diagnostics

The header bar displays live telemetry: Resident Set Size (RSS RAM), Virtual Memory Size (VSZ), network latency to peers, battery health, and thermal state throttling.

5. Troubleshooting & FAQ

Issue	Root Cause	Resolution
Windows app closes immediately on launch	Running mesh_ui.exe from inside the zip without extracting.	Right-click mesh_ui_windows_release.zip, select 'Extract All' , and run the extracted exe alongside its DLLs.
Missing MSVCP140 / VCRUNTIME140	Missing Visual C++ Redistributable on target PC.	Install the official Microsoft Visual C++ 2015-2022 Redistributable (x64) .
Peers not discovering each other	Devices on different subnets or firewall blocking mDNS / ports.	Ensure both devices are on the same Wi-Fi. Explicitly enter the local IP address (e.g. 172.20.10.4:50061) in the Next Peer field.
Installing on iPad / iPhone	iOS requires sideloading or development signing.	Use AltStore , Sideloadly , or Apple Configurator 2 to install Mesh_UI_Release.ipa onto your iPad.

MeshNode Decentralized AI Platform — Generated for Local Workspace Deployment